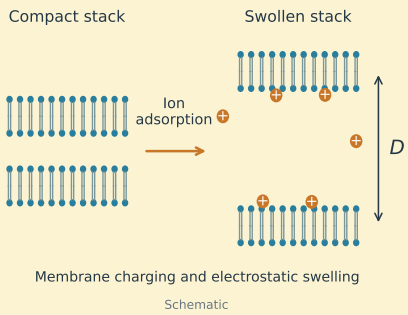
Ion-dependent lamellar organization of DPPC membranes: a SAXS study

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ABSTRACT: Multivalent ions can adsorb onto dipolar lipid membranes, charge their surfaces, and modify the interactions between adjacent bilayers. Using solution small-angle X-ray scattering, we investigated the interactions of Cu^{2+}, Co^{2+}, Ni^{2+}, Mn^{2+} and Fe^{3+} with 1,2-dipalmitoyl-*sn*-glycero-3-phosphocholine (DPPC) membranes over salt concentrations of 0.1–100 mM. At low concentrations, the lamellar repeat distances for most divalent ions were close to those of DPPC in water, suggesting very weak adsorption. At ion-dependent concentrations, scattering became dominated by the bilayer form factor, consistent with swelling of the lamellar stacks. We define the lower and upper concentration boundaries of this regime as C_1 and C_2. Above C_2, repeat distances generally decreased with increasing salt concentration. We interpret this behavior in terms of adsorption saturation and increased screening of electrostatic repulsion by ions remaining in the bulk solution. The concentration-dependent swelling and screening behavior reveals differences in the apparent affinity of the ions for DPPC and their effects on membrane organization.



INTRODUCTION

Phosphatidylcholine membranes have zwitterionic headgroups, yet their interactions with dissolved ions need not resemble those of an uncharged inert surface. Ion association, hydration and the local organization of headgroups can change membrane structure and the forces between adjacent bilayers. Work from the Raviv Group established that the structures of both the lipid and the ion affect the organization and effective charge of dipolar membranes [1]. The relevant experimental question is consequently more specific than whether a salt is monovalent or multivalent: which structural changes occur for a given lipid, salt, concentration and preparation history?

Lamellar small-angle X-ray scattering (SAXS) provides access to this question through correlation peaks and the bilayer scattering envelope. When successive orders can be assigned, their positions yield the lamellar repeat distance, D. This distance includes a bilayer and its intervening aqueous region (Figure 1); it is not a direct measurement of the amount of bound ion or of the water gap alone. Related Raviv Group studies used calcium concentration, equilibration and osmotic-stress comparisons to investigate adsorption onto dipolar membranes [2, 3]. Calcium–bicarbonate work further demonstrates that solution composition can matter alongside the nominal cation concentration [4]. These studies motivate a structural comparison while also defining the additional evidence required for a mechanistic interpretation.

Interactions between DPPC and the metal ions considered here have earlier experimental antecedents. Manganese has been used in electron-paramagnetic-resonance measurements of ion–phosphatidylcholine interactions [5]; cobalt and nickel have been examined by fluorescence quenching in DPPC bilayers [6]; and copper-containing DPPC systems have been studied using SAXS and anomalous SAXS [7]. Iron-binding measurements in liposomes of different phospholipid compositions provide a further, composition-dependent comparison [8]. These experiments differ in lipid formulation, thermal state, concentration or measurement principle. They preclude a claim that the individual metal–DPPC systems were previously unstudied, while leaving the present concentration series as a specific comparative experimental question.

Here we examine the SAXS signatures of DPPC dispersions prepared at a nominal lipid concentration of 30 mg/mL with five chloride salts: $CuCl_2$, $CoCl_2$, $NiCl_2$, $MnCl_2$ and $FeCl_3$. We compare lamellar repeat distances and the occurrence of identified correlation peaks across the concentration series, distinguishing structural evidence from interpretations in terms of ion association and screening. Alternative peak assignments are reported explicitly. The concentration-dependent swelling and screening behavior is used to compare the apparent interactions of the ions with DPPC membranes.

MATERIALS AND METHODS

Materials. DPPC (1,2-dipalmitoyl-*sn*-glycero-3-phosphocholine) was purchased from Avanti Polar Lipids (Alabaster, AL, USA) as lyophilized powder, with manufacturer-reported purity above 99%, and was used as received. Highly purified water was produced using a Barnstead Nanopure Diamond system. $CuCl_2$, $FeCl_3$, $MnCl_2$, $CoCl_2$ and $NiCl_2$ were purchased from Sigma-Aldrich (Rehovot, Israel), and absolute ethanol from Bio-Lab (Jerusalem, Israel). Salt concentrations denote the nominal added concentrations.

Sample preparation. DPPC powder was mixed with ethanol at 30 mg/mL. The solution was vortexed for approximately 10 min, heated in a dry-bath incubator at 45 °C for approximately 10 min, and sonicated for 10 min to achieve homogeneity. Aliquots of approximately 0.1 mL were distributed between Eppendorf tubes, frozen in liquid nitrogen and lyophilized overnight.

Salt solutions at concentrations between 0.1 and 100 mM were prepared in highly purified water and added to the dried lipid. The added volume, approximately 0.1 mL, was adjusted to obtain a final nominal DPPC concentration of 30 mg/mL. This preparation

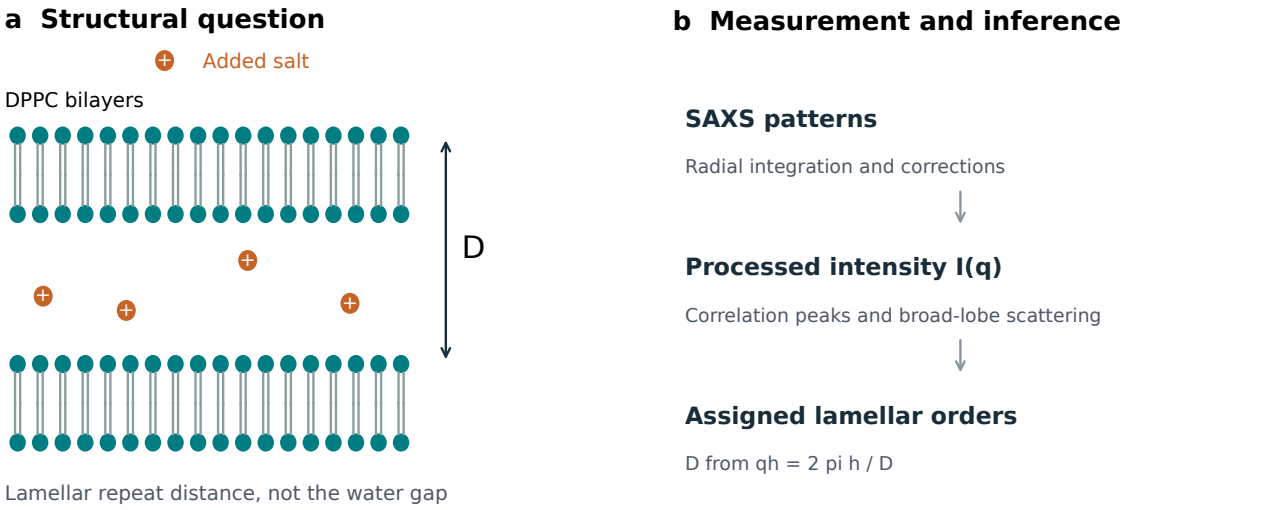


Figure 1. System and measurement. (a) Schematic DPPC bilayers in a salt-containing aqueous environment, defining the lamellar repeat distance D between corresponding positions in adjacent bilayers. Ion locations, relative dimensions and charge symbols are illustrative; the drawing does not assign adsorption sites or imply a measured ion distribution. (b) The measurement and analysis route, separating processed scattering from an assigned lamellar repeat distance.

concentration was kept fixed across the experiments. Under these conditions, the lipid formed multilamellar structures.

Three to five freeze–thaw cycles were applied to facilitate ion transfer across the bilayers through transient defects. In each cycle, the tubes were frozen in liquid nitrogen for 10 min and heated in a dry-bath incubator for 45 min. The samples were stored at 4 °C and vortexed for approximately 10 min before measurement preparation.

After approximately seven days, the dispersions were transferred to quartz capillaries. Some capillaries were sealed with Teflon tape and others were flame sealed. The tape-sealed capillaries were centrifuged for 10 min at 3000g using a Thermo Electron Corporation centrifuge; flame-sealed capillaries were centrifuged at 6000g using a SIGMA 1-15PK centrifuge with capillary rotor No. 11024. Samples were refrigerated for another one to two days before SAXS measurements. Background capillaries containing the corresponding salt solutions were also measured. Separate No Cycling and later centrifugation records are identified in the supporting tables.

SAXS analysis. Our in-house high-resolution solution SAXS setup was used as described elsewhere [9, 10]. The two-dimensional scattering patterns were radially integrated using FIT2D [11]. The scattering intensity, I, was analyzed as a function of the magnitude of the scattering vector,

$$q = \frac{4\pi}{\lambda} \sin\theta, \tag{1}$$

where 2θ is the scattering angle and λ the X-ray wavelength. Dark-current and salt-solution corrections were applied in the processing workflow; the numerical source and correction stage of each selected profile are documented in the Supporting Information.

The profiles were analyzed with X+, developed in our laboratory [12]. Bilayer form factors were described using electron-density models composed of Gaussian profiles normal to the membrane surface [10]. Lamellar correlation peaks were represented by Gaussian line shapes and indexed according to

$$q_h \simeq \frac{2\pi h}{D}, \tag{2}$$

where h is the assigned order and D the lamellar repeat distance. Single-peak assignments are marked as tentative. The plotted model curves are retained numerical fit outputs; the accompanying plotting scripts do not perform new nonlinear fits. Individual spacing assignments and alternatives are shown separately, without averaging. Complete source values, processing checks and supporting calculations are provided in the Supporting Information.

RESULTS

Scattering profiles and lamellar repeat distances. We investigated the interactions of transition-metal ions with DPPC membranes in solution using small-angle X-ray scattering (SAXS). The scattering profiles exhibited weak, moderate, strong or negligible structure-factor contributions. The form-factor contribution became more dominant as the lamellar structure-factor peaks weakened. Figure 2 shows two representative profiles: $MnCl_2$ at 20 mM, dominated by a broad bilayer envelope, and $NiCl_2$ at 100 mM, with pronounced lamellar correlations.

The structure-factor peaks were analyzed using X+, yielding the lamellar repeat distance, D. Multiple harmonically related peaks support indexing to a lamellar structure. The nickel profile contains two prominent features near 0.74 and 1.47 nm^{-1}, consistent with successive lamellar orders. The models describe the principal features of the profiles, with the data-minus-model differences shown below the corresponding curves. Complete-range diagnostics are provided in the Supporting Information.

Concentration-dependent membrane organization. Figure 3 presents the repeat distances as a function of salt concentration. Although the behavior depends on ion identity, the profiles can broadly be described in terms of compact lamellar stacks at low

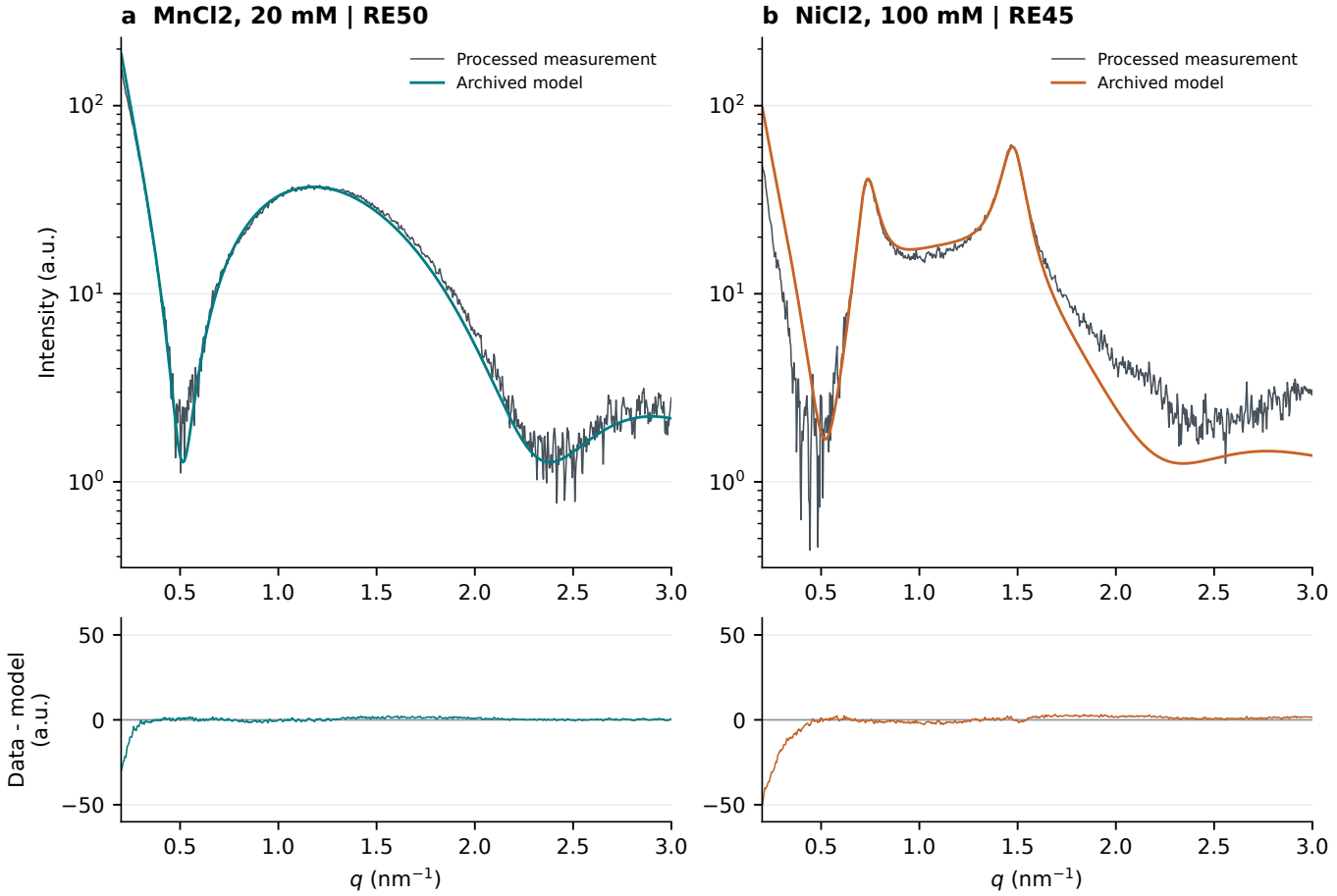


Figure 2. SAXS intensity and model curves for DPPC in (a) 20 mM $MnCl_2$ and (b) 100 mM $NiCl_2$. The manganese profile is dominated by the bilayer form factor; the nickel profile contains pronounced lamellar correlation peaks. Lower panels show data-minus-model differences. The displayed range is 0.2–3 nm^{-1}; complete profiles are included in the Supporting Information.

concentrations, strong swelling at intermediate concentrations, and decreasing repeat distances at higher concentrations. Conditions dominated by the bilayer form factor are considered separately from conditions with resolved lamellar peaks. The supporting tables retain the corresponding assignments and alternative values.

For Cu^{2+}, the repeat distances at 0.1 and 1 mM were close to those of DPPC in water, approximately 6.33–6.35 nm. The two 1 mM measurements gave 6.337 and 6.347 nm, consistent with finite lamellar spacing. At 10 mM, no lamellar peaks were assigned. This comparatively abrupt change differs from the more gradual swelling observed in parts of the other salt series. At 100 mM, the recorded value was approximately 13.4 nm, with an alternative fit of 18.4 nm; the alternatives are retained separately.

The Co^{2+} repeat distances at 0.1 and 1 mM were also near 6.34 nm. At 10 mM, scattering was assigned no lamellar peaks. At higher concentrations, the repeat distances decreased from approximately 14 nm at 30 mM to 12.35 nm at 40 mM, 11.65 nm at 50 mM and 8.96–9.01 nm at 100 mM.

For Ni^{2+}, the low-concentration distances were near 6.3 nm, with a weak single-peak assignment also recorded at 0.1 mM. At 10 mM, no lamellar peaks were assigned. At 20 mM, a repeat distance of 26.2 nm was observed, while another summary entry had no assigned distance. The repeat distances then decreased to 13.49, 11.95 and 10.45 nm at 30, 40 and 50 mM, respectively, reaching 8.52–8.56 nm at 100 mM.

For Mn^{2+}, the repeat distances at 0.1 mM were approximately 6.32–6.35 nm. At 1 mM, no lamellar peaks were assigned. At 10 mM, one profile had no assigned peaks, while another had a weak single-peak assignment corresponding to approximately 22.2 nm. The selected 20 mM profile was dominated by the bilayer form factor (Figure 2). At 30, 40 and 50 mM, the recorded distances were 15.79, 12.11 and 12.43 nm, respectively, followed by 9.66–9.68 nm at 100 mM. The 40-to-50 mM increase is an exception to the general high-concentration decrease.

For Fe^{3+}, low-concentration profiles included both unassigned peaks and tentative single-peak periodicities, with a large assigned distance near 23.9 nm at 1 mM. Distances at 10 and 20 mM were approximately 18–19 nm. At 30 and 40 mM, they decreased to 14.75 and 12.41 nm; the two labeled 50 mM values were 11.28 and 11.78 nm. At 100 mM, the recorded distances were 9.40 and 9.57 nm. The added iron salt was $FeCl_3$.

DISCUSSION

Ion adsorption and membrane swelling. The repeat-distance changes are interpreted in terms of ion adsorption onto otherwise dipolar DPPC membranes. Adsorbed ions charge the membranes, producing electrostatic repulsion between adjacent bilayers and increasing their separation. This interpretation follows the framework established for ion adsorption onto phosphatidylcholine membranes [1–3].

Below C_1, ion adsorption was inferred to be very weak. At 0.1 mM, the repeat distances for most divalent ions were nearly identical to those of DPPC in water, approximately 6.33 nm. Ni^{2+}

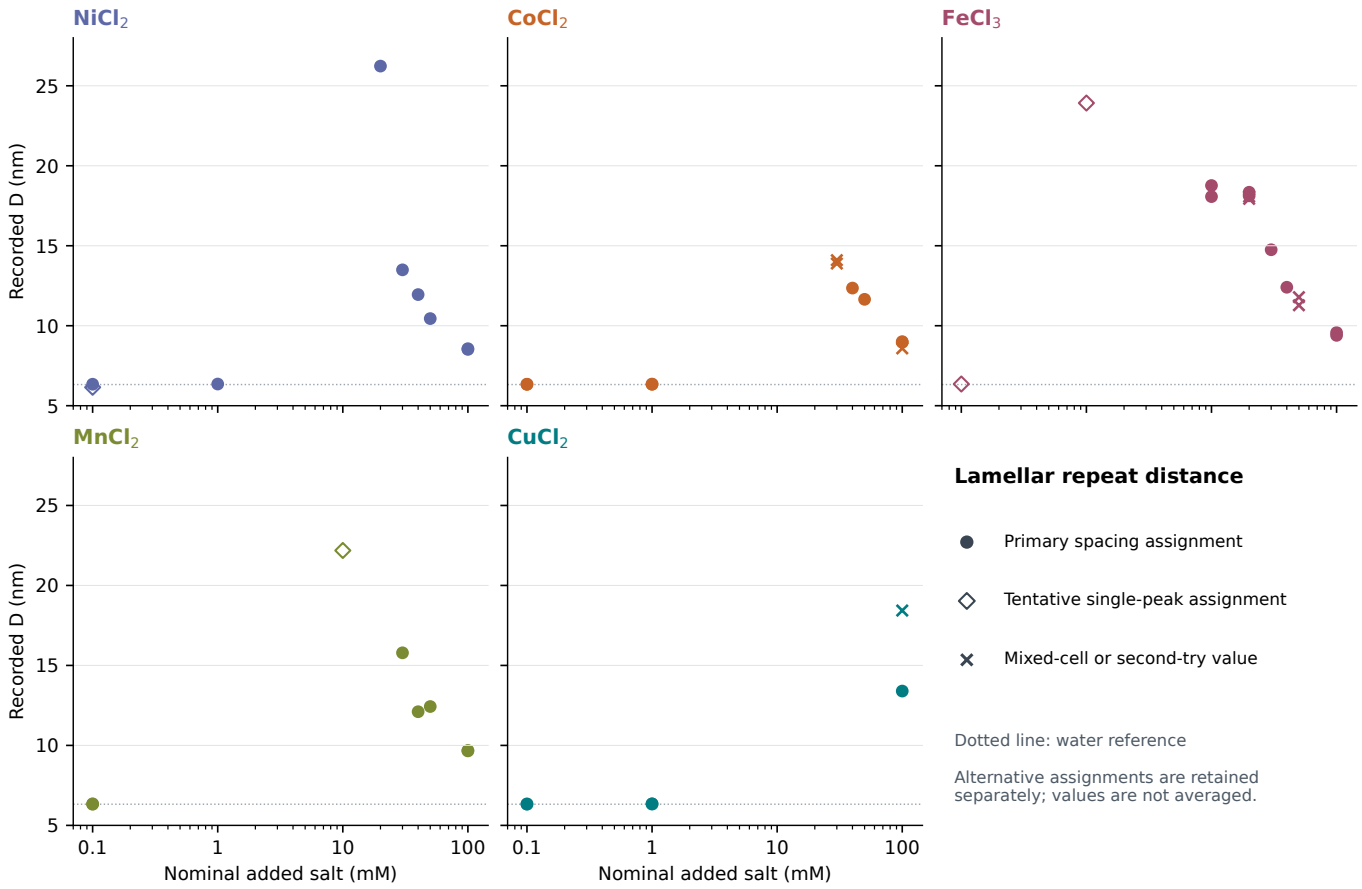


Figure 3. Lamellar repeat distance, D, of DPPC as a function of added salt concentration. The panels follow the qualitative affinity order discussed in the text: Ni^{2+}, Co^{2+}, Fe^{3+}, Mn^{2+} and Cu^{2+}. Solid circles denote primary numeric assignments; open diamonds denote tentative single-peak assignments; crosses denote mixed-cell or second-try values. The dotted line indicates the recorded water reference. Alternative assignments are shown separately, without averaging. Complete values and source assignments are provided in the Supporting Information.

showed a small deviation, while Fe^{3+} exhibited a different scattering response. Apart from Fe^{3+}, these results suggest very little ion adsorption onto DPPC membranes up to 0.1 mM.

For Cu^{2+}, the repeat distances at both 0.1 and 1 mM remained close to the water value, suggesting weak adsorption over this concentration range. At 10 mM, scattering was dominated by the bilayer form factor, which we interpret as substantial swelling associated with adsorption and membrane charging. For Co^{2+}, the more limited swelling regime and relatively short repeat distances at higher concentrations suggest weaker adsorption than for some of the other ions.

Critical concentrations, saturation and screening. We define two critical salt concentrations for the form-factor-dominated regime: C_1 is the lowest sampled concentration at which a pure form-factor scattering profile was observed, and C_2 is the highest. These are operational boundaries of the observed concentration regime. The loss of resolvable lamellar peaks suggests a large separation between bilayers, which we attribute to the electrostatic repulsion generated by ion adsorption.

Above C_2, the repeat distances generally decreased with increasing ion concentration. We attribute this decrease to an increasing concentration of ions outside the multilamellar phase. In the proposed saturation-and-screening picture, as the bilayers approach saturation with adsorbed ions, additional ions remain predominantly in the bulk solution. These ions screen the electrostatic repulsion between the charged bilayers, reducing the lamellar repeat distance. Adsorption saturation is part of this physical interpretation, rather than a separately measured occupancy plateau. Increasing screening can also reduce the spacing while additional adsorption continues [1].

The large repeat distance observed with 20 mM $NiCl_2$ is consistent with weak screening in this part of the concentration range. At higher concentrations, the nickel repeat distance decreased monotonically. For $MnCl_2$, the large, tentative 10 mM assignment likewise suggests weak screening; the form-factor-dominated 20 mM profile illustrates the nonuniform behavior within this series. At 100 mM, the Co, Ni, Mn and Fe distances had decreased to approximately 8.5–9.7 nm. The copper series remained different, with a much larger recorded spacing. We retain the Mn 40–50 mM reversal as an observed exception without assigning it to a quantified experimental scatter.

Relative ion affinity. On the basis of the concentration-dependent swelling and screening behavior, we infer the qualitative relative affinity of the ions for DPPC to follow

$$Ni^{2+} < Co^{2+} < Fe^{3+} < Mn^{2+} < Cu^{2+}. \quad (3)$$

The concentration comparisons in Figure 3 therefore follow the order nickel, cobalt, iron, manganese and copper. This ordering expresses the interpretation of the measured structural response under the conditions of this study. It is a qualitative membrane-affinity comparison, not a set of independently measured binding constants. Comparable inferences from ion-dependent swelling

have been made for other phosphatidylcholine systems [1].

The comparison involves both adsorption-induced charging and the screening contributed by free ions. In particular, $FeCl_3$ and the divalent chlorides do not have equal ionic strength at the same nominal molarity. Ion valence, solution composition and lipid structure can consequently contribute to differences between the curves [1, 4, 13]. Earlier DPPC studies using EPR, fluorescence quenching and anomalous X-ray scattering provide complementary evidence for metal–membrane interactions [5–7]. The affinity order in eq 3 concerns adsorption of aqueous ions to membranes; it is not identified with the electron affinity of isolated atoms.

CONCLUSION

The interaction of transition-metal ions with DPPC membranes depends strongly on ion identity and concentration. At low concentrations, water-like repeat distances suggest weak adsorption for most divalent ions. At intermediate concentrations, adsorption-induced charging is associated with strong swelling and form-factor-dominated scattering. At higher concentrations, the repeat distance generally decreases, which we interpret in terms of adsorption saturation and increased screening by ions in the bulk solution. These observations provide a qualitative comparison of the ions' affinity for DPPC and show how ion adsorption and screening regulate lamellar membrane organization.

AUTHOR CONTRIBUTIONS

Roee Tekoah performed sample preparation, SAXS measurements, integration and background subtraction, X+ fitting, figure preparation, interpretation and manuscript preparation. The research was conducted in the Raviv Group at the Hebrew University of Jerusalem.

DATA AND CODE AVAILABILITY

The accompanying SaLSA project repository contains processed scattering curves, source tables, sample and processing links, plotting and descriptive-analysis scripts, numerical results, figures and the paper PDF. The scripts reproduce the figures and selected calculations from the processed inputs. They do not re-integrate detector frames or rerun the nonlinear X+ fits. Competing source assignments and processing differences are documented with the data.

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SUPPORTING INFORMATION

Ion-dependent lamellar organization of DPPC membranes: a SAXS study

Concentration-series assignments

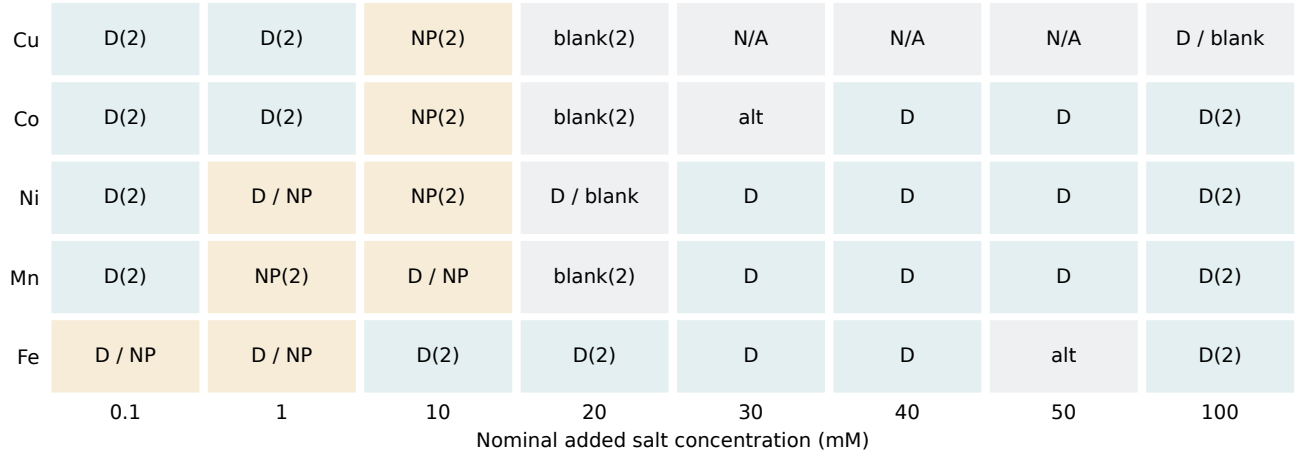


Figure S1. Status of main-summary cells across the nominal condition grid. D denotes a numeric spacing cell, NP the historical "No Picks" label, alt a cell containing multiple or alternative values, blank an empty spacing cell, and N/A a recorded unavailable result. Parentheses count source rows rather than independently prepared samples. D includes provisional single-peak assignments distinguished in Figure 3. This is an observation-record map, not a phase diagram. No Cycling entries are kept separately in the supplement.

Full-range model diagnostics

Each original export contains 1621 numerical rows over $q = 0.070376$–5.3731 nm^{-1}. Figure 2 displays 0.2–3 nm^{-1}, where all displayed intensities are positive. Figure S2 shows the complete range on a symmetric logarithmic scale, retaining the 39 nonpositive manganese values and 433 nonpositive nickel values. Substantial low-q departures occur below the main display window. Without the original fit intervals, weights and error model, these differences are descriptive and are not reported as reduced chi-square values.

Illustrative background-factor perturbation

At the archived subtraction factor, the two interpolated nickel peak positions are approximately 0.736219 and 1.467204 nm^{-1}, with a ratio of 1.99289. Assigning these features to orders one and two gives $D = 8.55874$ nm using the stated origin-constrained slope. The illustrative $\pm 1\%$ and $\pm 5\%$ factor changes produce estimates between 8.55858 and 8.55891 nm (Figure S3). Thus the positions of these two selected local maxima change little over this particular perturbation range, even though peak intensities change visibly. The discrete maximum bins remain unchanged; the small reported shifts come from sub-grid interpolation and do not estimate q precision.

The simple estimate differs from the recorded X+ value, 8.520042 nm, by approximately 0.0387 nm (0.45%). The recovered Ni parameter file declares two active peaks with stored centers 0.728879 and 1.479209 nm^{-1}. Applying the same origin-constrained indexing relation to those two centers gives 8.52004233 nm, reproducing the recorded distance to its printed precision. A leftover third-center key is inactive and is excluded. Thus the indexed historical value is arithmetically traceable to stored profile-fit centers, while the new calculation uses maxima of the measured intensity. The estimators need not agree when the envelope, background or peak asymmetry shifts the intensity maximum. The complete original fit objective and export-generating session remain unresolved. The perturbation calculation supports a narrow statement about two selected maxima under one processing change; it supplies neither an uncertainty bar nor a new validated X+ replacement.

Factor change	Factor	q_1 / nm^{-1}	q_2 / nm^{-1}	D / nm
-5%	1.509702	0.736136	1.467207	8.558913
-1%	1.573268	0.736203	1.467204	8.558771
+0%	1.589160	0.736219	1.467204	8.558737
+1%	1.605052	0.736234	1.467203	8.558704
+5%	1.668618	0.736294	1.467200	8.558578

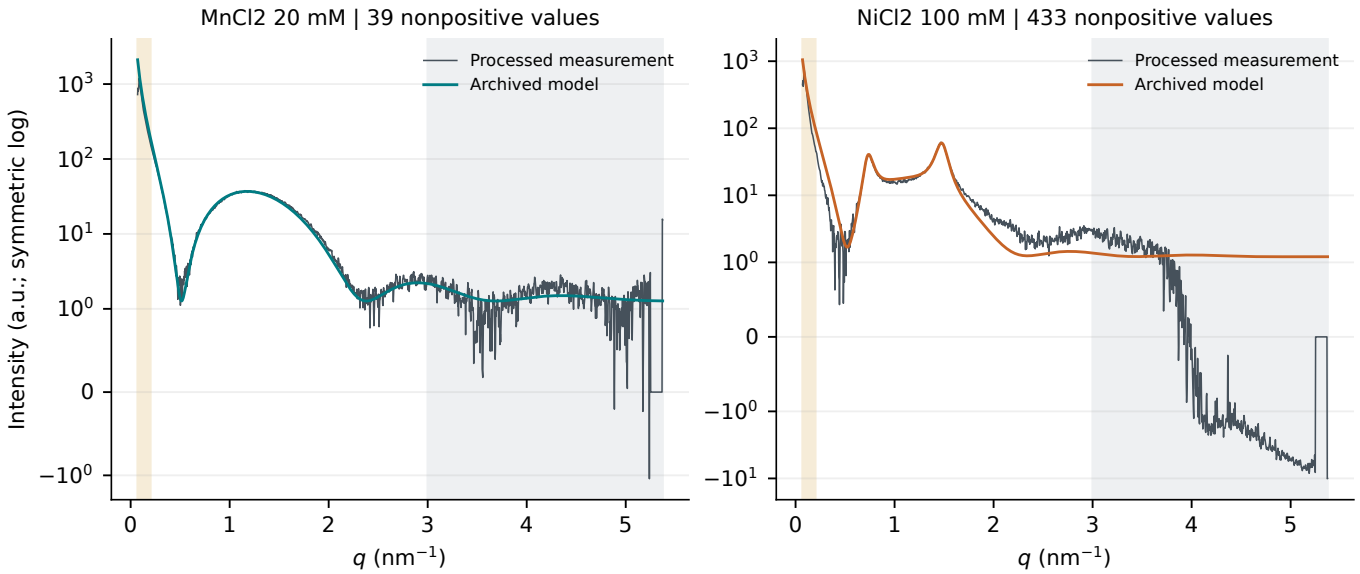


Figure S2. Complete exported range for the two representative curves. The symmetric logarithmic axis retains positive, zero and negative values, with a linear region around zero. Shading identifies $q < 0.2$ nm^{-1} and $q > 3$ nm^{-1}, outside the main display. These panels expose the low-q model departures and nonpositive high-q values rather than deleting them. Nonpositive values are properties of the processed output, not negative physical scattering cross-sections.

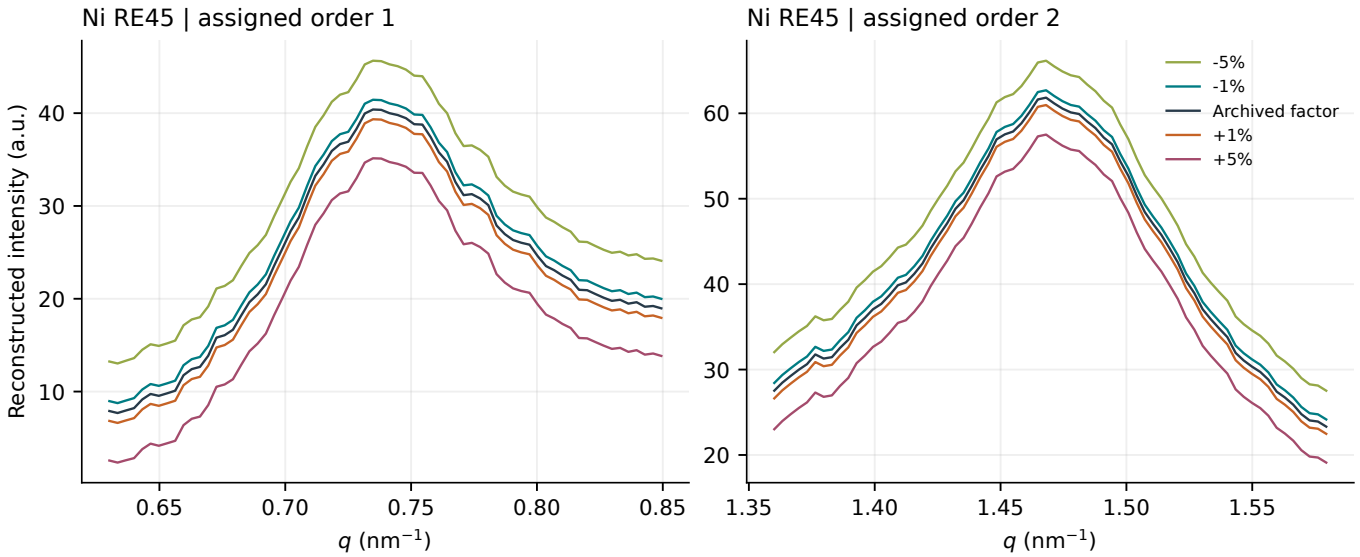


Figure S3. Reconstructed Ni RE45 peak regions at the archived background factor and illustrative ±1% and ±5% relative changes. Peak intensities respond to subtraction while the selected local maxima shift little over these particular perturbations. Percentages do not represent calibrated uncertainty. The order assignments and fixed windows are stipulated for this calculation; changing them, the raw integrations or the profile model was not tested.

Processing records and numerical checks

The experimental protocol uses the Raviv Group's in-house high-resolution solution SAXS setup and radial integration using FIT2D, followed by X+ analysis [11, 12]. Published descriptions of the group's setup and models provide methodological context [9, 10]; their instrument settings are not substituted for missing settings in these particular runs. The archive contains detector frames, acquisition logs, masked and unmasked integrations, dark-current-corrected outputs and, for some samples, salt-background-subtracted outputs. The present figures use selected processed outputs; they do not represent a fresh integration of detector frames.

For the traced Mn RE50 (20 mM) and Ni RE45 (100 mM) examples, the exported signals match their salt-background-subtracted numerical sources. Their subtraction arithmetic is

$$I_{\mathrm{sub}}(q) = I_{\mathrm{sample,DC}}(q) - \alpha I_{\mathrm{salt,DC}}(q), \tag{4}$$

with recorded factors $\alpha = 1.52781$ and 1.58916, respectively. Interpolation of the two input curves onto the output grid, without extrapolation, reproduces the archived subtraction to the precision of the printed exports. This verifies the arithmetic association; it does not independently validate the scaling factors, absolute intensity normalization or preceding dark-current corrections. Other examples do not necessarily use this same processing stage: the historical Fe 30 mM paper export matches a dark-current-corrected output, and is not used as a directly comparable overlay here.

The selected RE45 and RE50 logs have 7200 s filenames and 7201 s shutter intervals, but exposure fields near 3600.02 s. A note in the archived processing script documents a known wrap of logged durations for exposures exceeding one hour, requiring manual correction. That convention explains why the header alone cannot be used as the full exposure duration; the particular correction branch used for these samples has not been verified. Zero temperature fields accompanied by an oven-off note do not establish a temperature of 0 °C. The available records do not establish sample temperature, calibration geometry, resolution or normalization sufficiently to assign a membrane thermal phase.

Repeat-distance assignments and summary records

We use the scattering-vector convention

$$q = \frac{4\pi}{\lambda}\sin\theta, \tag{5}$$

where 2θ is the scattering angle and λ is the X-ray wavelength. The numerical curve axis is expressed in nm^{-1}, following the historical analysis. For a lamellar harmonic series with assigned order h,

$$q_h \simeq \frac{2\pi h}{D}. \tag{6}$$

Multiple consistent orders support the repeat-distance assignment. A single isolated feature yields $2\pi/q_*$ only after assigning it to the first order; without further evidence, this is a provisional periodicity rather than a demonstrated lamellar series. Single-peak flags in the historical tables are preserved in the figure.

The original analysis describes bilayer electron-density models built from Gaussian profiles and lamellar peaks represented by Gaussian line shapes. The published X+ framework supplies their broader modeling context [10, 12]. The model lines shown here are archived numerical exports, replotted without optimizing parameters. Peak widths, finite stacking, disorder and the bilayer envelope can affect the observed intensity; a broad envelope together with weak peaks is not by itself evidence for two distinct populations.

Three compact historical workbooks yield 72 source rows: 66 main-summary rows, including one water-control entry, and six No Cycling rows. Source rows, sample labels, exposures and alternative fits are separate entities. Neither these row counts nor matching sample labels establish a count of independently prepared samples. Empty cells, "No Picks", "N/A" and mixed alternative values are retained as different statuses. The historical Error column contains numbers, zero and text with an incomplete definition. It is preserved in the registry and supplement but is not used as an experimental uncertainty estimate or as a weight in the new calculations. No averaging or error bars are introduced.

A bounded background-scaling calculation

To examine one processing question, we reconstructed the Ni RE45 subtraction at relative factor changes of −5%, −1%, 0, +1% and +5%. These are illustrative perturbations, not an estimated distribution of scaling uncertainty. In fixed windows 0.63–0.85 and 1.36–1.58 nm^{-1}, chosen around the two visible peaks, we interpolated each local maximum using the three nearest points and a quadratic polynomial. With orders $h = 1, 2$ assigned, an origin-constrained least-squares slope $b = \sum h q_h / \sum h^2$ gives the descriptive estimate $D = 2\pi/b$. It does not reproduce the original X+ fit, test alternative order assignments, or estimate sample-to-sample variation.

Disposition of known conflicts

The representative paper export labeled Co 0.1 mM is excluded from the overlays because it matches the Cu RE3 numerical source. Its adjacent model is not reused under a corrected identity. Cu 1 mM is treated as finite-spacing evidence, supported by the recorded table and archived peak fits. The Fe 30 mM export is retained in the archive but excluded from the matched salt-background-subtracted overlay pair because its recovered source is DC-subtracted. Later Mn 30 mM fit editions disagree; the 15.788546 nm summary value is displayed as recorded, without choosing a conflicting session silently. N/A Cu cells are not converted to absent measurements, and blank entries are not converted to pure-form-factor labels. Source assignments and processing differences are documented in the accompanying data tables.

Complete historical source-row table

The following table preserves all 72 registry rows, including the separate No Cycling section. Row labels WB1/R2, for example, denote workbook 1, sheet 1, row 2; the CSV contains the complete workbook names, source hashes, raw notes, flags and processing status. "?" means the summary row does not provide a sample ID. Empty cells remain empty, not peak-free. Numeric digits reproduce the archive's printed values; they do not imply validated accuracy. The Error column is historical and is not used here as an uncertainty estimate. "Ambi" abbreviates the recorded "One Ambi Pick" wording. Mixed primary cells and second-try values are printed explicitly. Independent preparations remain unresolved.

Row	Label	Ion	*c*/mM	Spacing cell / nm	Historical Error
WB1/R2	RE 1	Cu	10	No Picks	
WB1/R3	RE 2	Cu	1	6.337113	0.001826
WB1/R4	RE 3	Cu	0.1	6.336852	0.001896
WB1/R5	RE 4	Cu	10	No Picks	
WB1/R6	RE 5	Cu	1	6.347240	0.002315
WB1/R7	RE 6	Cu	0.1	6.330824	0.001597
WB1/R8	RE 7	Co	10	No Picks	
WB1/R9	RE 8	Co	1	6.351310	0.001652
WB1/R10	RE 9	Co	0.1	6.345014	0.001747
WB1/R11	RE 10	Co	10	No Picks	
WB1/R12	RE 11	Co	1	6.338146	0.001458
WB1/R13	RE 12	Co	0.1	6.334378	0.001684
WB1/R14	RE 13	Ni	10	No Picks	
WB1/R15	RE 14	Ni	1	6.354941	0.001726
WB1/R16	RE 15	Ni	0.1	6.342976	0.001854
WB1/R17	RE 16	Ni	10	No Picks	
WB1/R18	RE 17	Ni	1	No Picks	
WB1/R19	RE 18	Ni	0.1	6.151151	0
WB1/R20	RE 19	Mn	10	No Picks	
WB1/R21	RE 20	Mn	1	No Picks	
WB1/R22	RE 21	Mn	0.1	6.347511	0.001724
WB1/R23	RE 22	Mn	10	22.185527	Ambi
WB1/R24	RE 23	Mn	1	No Picks	
WB1/R25	RE 24	Mn	0.1	6.323221	0.001555
WB1/R26	RE 25	Fe	10	18.071810	0.033856
WB1/R27	RE 26	Fe	1	No Picks	
WB1/R28	RE 27	Fe	0.1	6.356565	Ambi
WB1/R29	RE 28	Fe	10	18.764685	0.053829
WB1/R30	RE 29	Fe	1	23.922176	Ambi
WB1/R31	RE 30	Fe	0.1	No Picks	
WB1/R32	RE 31	control	–	6.323221	0.001555
WB1/R36	RE 1 (NC)	Cu	10	Form Factor+	
WB1/R37	RE 2 (NC)	Cu	1	6.343700	0.001239
WB1/R38	RE 3 (NC)	Cu	0.1	6.344891	0.001794
WB1/R39	RE 16 (NC)	Ni	10	Form Factor+	
WB1/R40	RE 17 (NC)	Ni	1	Form Factor	
WB1/R41	RE 18 (NC)	Ni	0.1	Form Factor+	
WB2/R2	RE 32	Fe	20	18.344936; second: 17.932439	0.015572
WB2/R3	RE 33	Fe	100	9.397259	0.007554
WB2/R4	RE 34	Fe	20	18.120405; second: 18.072541	0.051384
WB2/R5	RE 35	Fe	100	9.570564	0.002293
WB2/R6	RE 36	Cu	20		
WB2/R7	RE 37	Cu	100		
WB2/R8	RE 38	Cu	20		
WB2/R9	RE 39	Cu	100	13.397393; second: 18.431046	0.017374
WB2/R10	RE 40	Co	20		
WB2/R11	RE 41	Co	100	9.005215; second: 8.584269	0.011161
WB2/R12	RE 42	Co	20		
WB2/R13	RE 43	Co	100	8.964354	0.00826
WB2/R14	RE 44	Ni	20		
WB2/R15	RE 45	Ni	100	8.520042	0.00848
WB2/R16	RE 46	Ni	20	26.231220	0.031042
WB2/R17	RE 47	Ni	100	8.563865	0.009333
WB2/R18	RE 48	Mn	20		
WB2/R19	RE 49	Mn	100	9.681803	0.002539
WB2/R20	RE 50	Mn	20		
WB2/R21	RE 51	Mn	100	9.656939	0.006879
WB3/R2	?	Fe	30	14.749398	0.034498
WB3/R3	?	Fe	40	12.405140	0.019958
WB3/R4	?	Fe	50	RT56: 11.279192, RT57: 11.784526	0.062461
WB3/R5	?	Cu	30	N/A	N/A
WB3/R6	?	Cu	40	N/A	N/A
WB3/R7	?	Cu	50	N/A	N/A
WB3/R8	?	Co	30	2 peaks: 13.89, 3 peaks: 14.097209	0.014989
WB3/R9	?	Co	40	12.354106	0.002383
WB3/R10	?	Co	50	11.650784	0.003858
WB3/R11	?	Ni	30	13.493705	0.005144
WB3/R12	?	Ni	40	11.947227	0.005983
WB3/R13	?	Ni	50	10.446840	0.023937
WB3/R14	?	Mn	30	15.788546	0.015518
WB3/R15	?	Mn	40	12.106260	0.012808
WB3/R16	?	Mn	50	12.431007	0.010267